Respiratory Sound Analysis for Detection of Pulmonary Diseases

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***Abstract*— Large number of people die every year of Pulmonary chronic lung diseases irrespective of their age. Lung sound analysis has been a key diagnostic aid to accurately detect Pulmonary Diseases. Earlier, manual detection was used which was not a dependable method to detect lung diseases due to various reasons like low audibility and difference in perceptions of different physicians for different sounds. Modern computerized analysis yield results with much higher accuracy and thus a better treatment can be given to patients suffering from various kinds of lung diseases. These disorders include Asthma, Bronchitis, Emphysema, Tuberculosis and Pneumonia. Some of the symptoms are wheezing, shortness of breath, rhonchi and chronic cough. In general, the analysis is carried out in two stages- Separation of Heart Sound (HS) from the Lung Sound (LS) and the characterization of waveform of the filtered LS. In this paper, we propose a very simple yet effective method for the second stage analysis- characterization of waveform of the filtered LS for some of the male and female age groups. We have taken the filtered Lung sounds from different online repositories and performed Welch method. This method helps to obtain power spectrum plot of a particular LS. Different diseases have peaks in different frequency ranges of the power spectrum plot. This helps in identification of a particular disease.**

Keywords—Pulmonary diseases, Respiratory sounds, Welch method.

I. INTRODUCTION

Pulmonary disorder is the inability of a person to breathe normally. Manual analysis used in the past only gave an approximate idea of the disorder and hence a very rough treatment was given. This was working out well in the past. Drastic increase in pollution and non-healthy habits of people has given rise to more complex diseases and need a very accurate estimation of the extent of disease. This accuracy can only be bought by automation of the analysis. Researchers observed that the difference between sounds made by infected lungs and the normal healthy lungs could serve as a very good tool for the detailed study and detection of the disease. Recording the Lung sounds, filtering them from the Heart sounds and other noises and studying the waveform of the filtered Lung sound has been the *de facto* way of performing the analysis. Many methods are given for the filtering and processing of the Lung sounds.

Brief review of the previous papers reveals several methods of filtering and examining the LS. The most challenging task in the analysis is the separation of HS from the LS due to spectral and temporal overlap between the two sounds**.** Filtering techniques used are Modulation Domain filtering [5] that filters the temporal trajectories of short-term spectral components.

Signal analysis is carried out by segmentation into consecutive overlapping frames and performing Fourier transform. Combination of adaptive-frequency domain filtering [10] where a very simple method is described that involves subtracting heart sounds from combination of heart and lung sound. Use of Kaiser window, FIR band-pass filter [6] using LPF, differentiators and Hilbert transformers. Usage of combination of BPF, LPF and HPF adaptive filters [7]. However due to spectral and temporal overlap of the HS and LS, the complete separation of the two sounds is very difficult. On conducting a brief study of previous papers, we found that there are different ways proposed which have some traces of HS. After the filtering, the pure LS is sent for further processing. Benchmark Separation algorithm for wavelet-based analysis [5] for characterization of LS is used. Machine learning algorithm Back-Propagation Neural Network [7] is used to match the recorded waveform from the standardized waveforms of different diseases. Lung sounds are collected using piezoelectric transducer and analyzed using power density spectrum [11]. A first-hand diagnosis tool is made for lung sound analysis [12] where results are obtained based on different scans like CT scan and X-ray.

II. METHODOLOGY

Various types of infected lung recordings are taken for different age groups from online repositories [1][2][3][4]. The method detects and characterizes the Lung sound waveforms in MATLAB. Various types of Lung sound waveforms that are detected in Welch method are wheezes, crackles, coarse crackles and stridor. The method is demonstrated on the lung sounds of a normal human being. The same is performed on affected lungs and results are shown in the results section.

Welch method:

1. Cycle Extractor: The input signal let it be x[n] (filtered lung sound) is broken down into L overlapping parts $x_l[n]$ each of length N, where l is in the range [0, L-1] with step size M in [0, N]. Window is applied to each of L data parts (in the time domain) which gives l-th windowed segment. This extraction helps in separating out one dilation and constriction of lungs that is used for further analysis. The windowing of the parts is what makes the Welch method a modified periodogram as shown in fig 1.

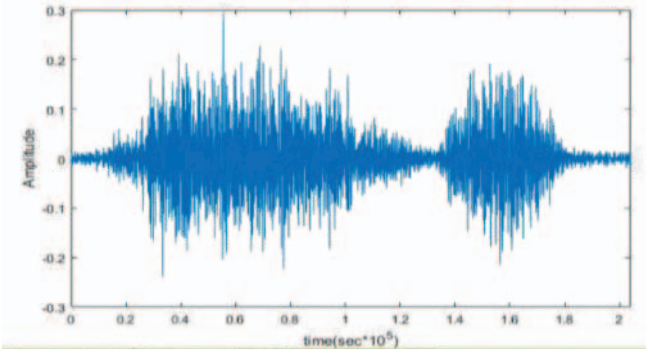


Fig. 1. Amplitude-time graph of single cycle of normal Vesicular LS.

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2. FFT block: In this step, Discrete Fourier transform of the extracted l-th windowed segment is calculated using (1). Resulting FFT spectrum is shown in fig 2.

$$X_l(e^{j\omega}) = \sum_{n=0}^{N-1} x[n+l\cdot M]\cdot w[n]\cdot e^{-j\omega n} \qquad (1)$$

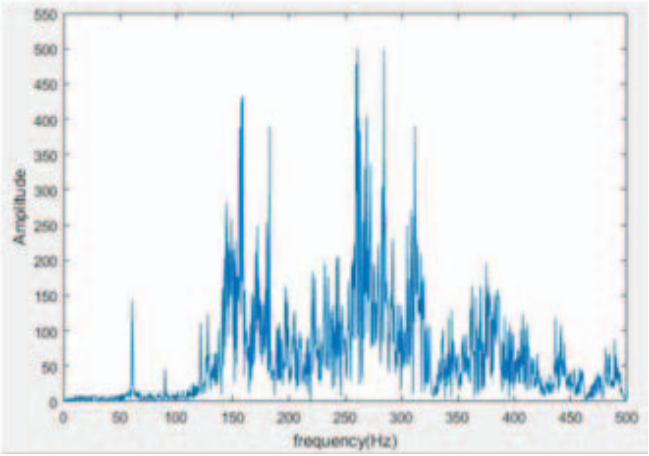


Fig. 2. FFT plot of single cycle of normal Vesicular LS.

3. Power spectrum block: Square magnitude of the resulting l-th periodogram obtained from previous step and is calculated by the formula:

$$S_{xx,l}(e^{\omega j}) = \frac{1}{N} |\, X_l(e^{\omega j})\,|^2 \qquad (2)$$

The discrete periodogram are then averaged, which lessens the variance of the distinct power measurements. The final result is an array of power measurements vs. frequency "bin" as shown in fig 3. The formula for averaging is:

$$S_{xx}(e^{\omega j}) = \frac{1}{L}\sum_{l=0}^{L-1} S_{xx,l}(e^{\omega j}) \qquad (3)$$

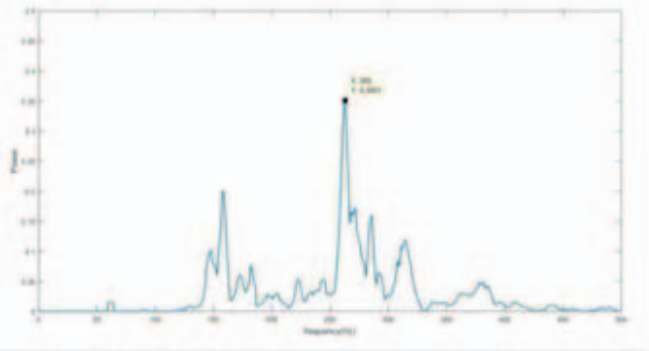


Fig. 3. Power spectrum of normal Vesicular LS by welch method.

4. Detector: The peak from the power spectrum is observed and the frequency range in which the peak frequency lies is noted. This range is then compared with standard dataset of frequency ranges specified for each disease for different age groups. If the peak lies between 200-250Hz, the patient is normal else the patient is suffering from some lung disorder depending on the peak in the power spectrum.

There are different specified frequency ranges for different types of disorders like asthmatic lungs (150-200Hz) [8] and fine crackles (50-100Hz). Frequency of all lung sounds lie in the range of 50-2500Hz [9].

In the methodology, we have taken Normal Vesicular Lung sound and performed the Welch method. A brief block diagram of the proposed algorithm is shown in figure 4.

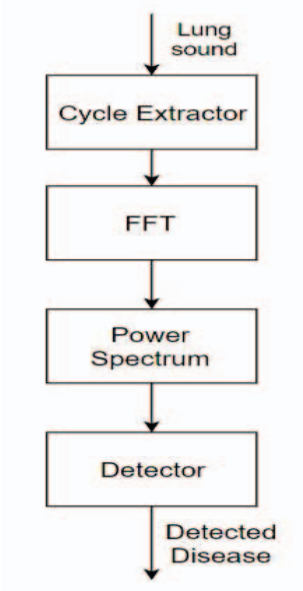


Fig. 4. Block Diagram of Welch Method.

III. Results And Discussions

Three types of infected lung sounds are taken namely wheezes, fine crackles and stridor. Different sounds gave peaks in different frequency ranges in the power spectrum plot after passing through the series of blocks shown in Fig 4. Infected lungs are differentiated from normal lungs by the peak that is obtained in different frequency ranges in the power spectrum plot of the respective sound.

In the past, people have worked upon the Welch method but conducted different analysis and have not used it for disease detection. People have segregated lung sounds as diseased or good but not to identify type of disease. The novelty of our work comes from its unique application. The lung sounds taken from the online repositories mentioned the symptoms and disease each patient had. We took all the lung sounds for a particular disease and obtained the frequency range of its peak. Table 1 tabulates our finding of frequency ranges for different subject symptoms, age and gender.

First set of inputs are the filtered Lung sounds of symptom “Wheezes”. These inputs are passed through Welch block diagram in Fig 4. Under Wheeze symptom, we found that a teenage male with lung sound peak frequency range 150-200Hz (Fig 5) probably has Pulmonary Edema.

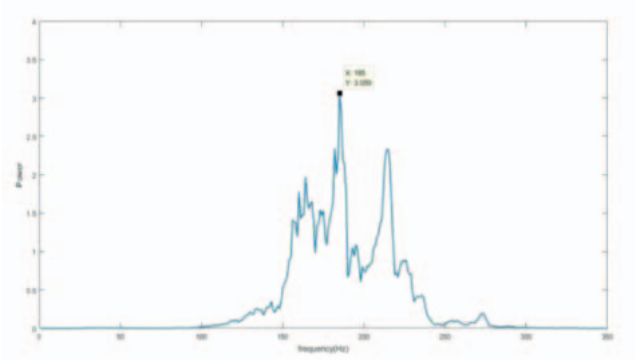


Fig. 5. 18-year-old person with pulmonary edema.

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A male child (4-13years) with lung sound peak frequency range 0-50Hz (Fig 6) probably has Asthma.

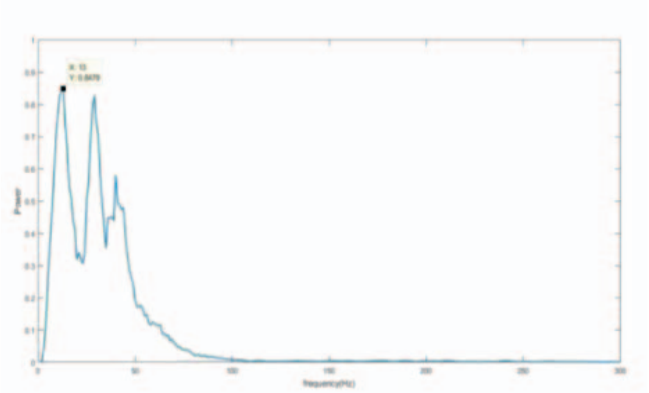


Fig. 6. 8-year-old boy with asthma.

A male infant with lung sound peak frequency range 200-250Hz (Fig 7) probably has Viral Bronchitis.

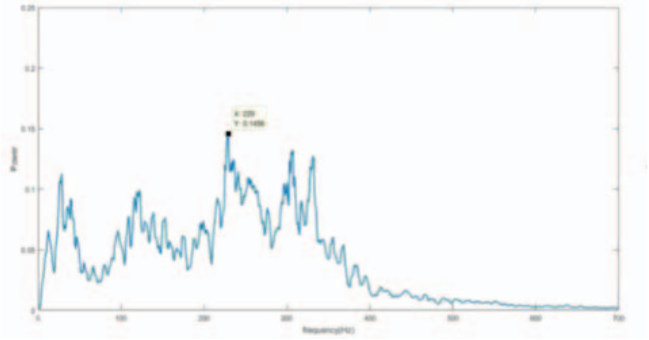


Fig. 7. 8-month-old boy with viral bronchiolitis.

A female child (4-13years) with lung sound peak frequency range 0-50Hz (Fig 8) probably has Acute Asthma.

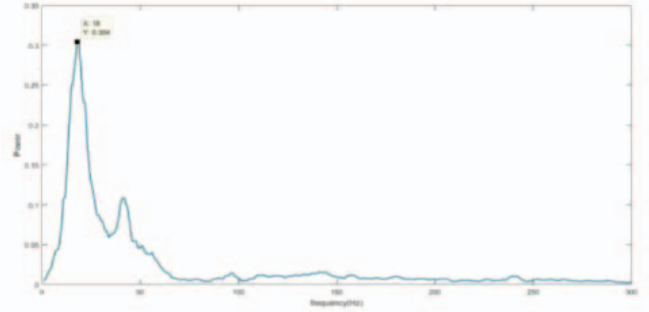


Fig. 8. 11-year-old girl with acute asthma.

Second set of inputs are the filtered Lung sounds of symptom “fine crackles”. These inputs are passed through Welch block diagram in Fig 4. A teenage male with lung sound peak frequency range 0-50Hz (Fig 9) probably has tuberculosis

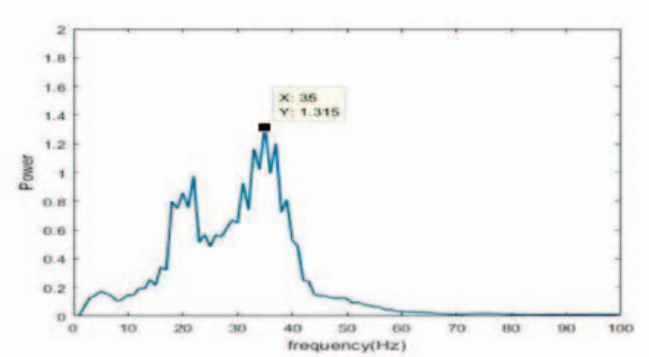


Fig. 9. 16-year-old boy with tuberculosis.

An old woman (>50 years) with lung sound peak frequency range 0-50Hz (Fig 10) probably has Rheumatoid.

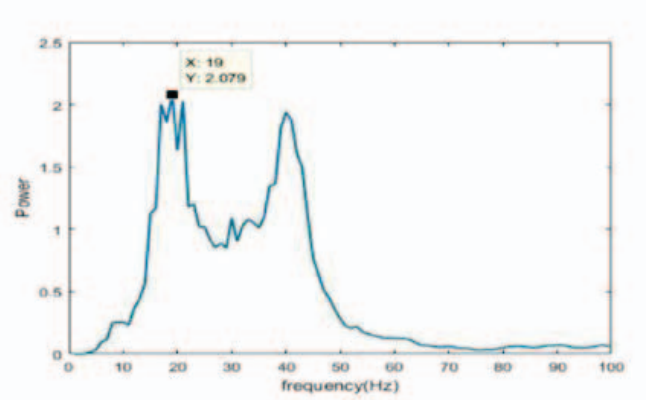


Fig. 10. 55-year-old woman with rheumatoid lung disease.

A male child (4-13years) with lung sound peak frequency range 250-350Hz (Fig 11) probably has Pneumonia.

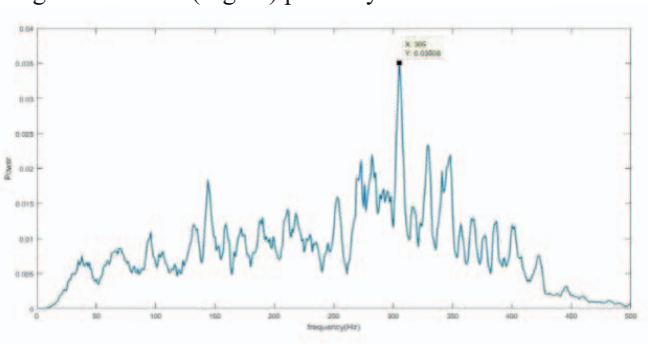


Fig. 11. 9-year-old boy with pneumonia.

Third set of inputs are the filtered Lung sounds of symptom “stridor”. These inputs are passed through Welch block diagram in Fig 4. A male infant with lung sound peak frequency range 800-900Hz (Fig 12) probably has Epiglottis.

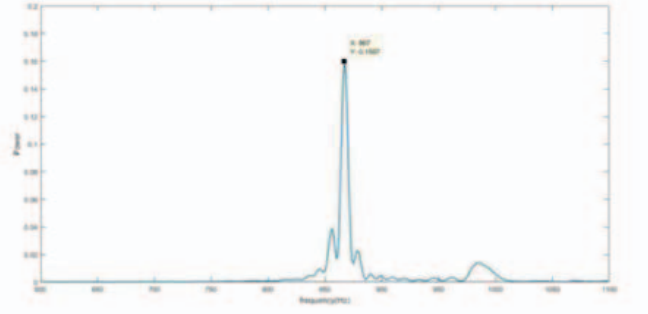


Fig. 12. 7 months old with epiglottitis.

A female infant with lung sound peak frequency range 650-750Hz (Fig 13) probably has Laryngomalacia.

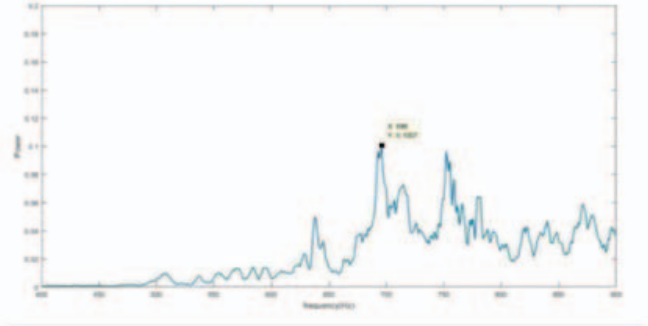


Fig. 13. Laryngomalacia in Infant.

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TABLE I RESULTS OBTAINED

Symptoms	Age	Frequency range	Gender	Disease
Wheezes	Teenage	150-200Hz	Male	Pulmonary Edema
	Child (4-13 years)	0-50Hz	Male	Asthma
	Infant	200-250Hz	Male	Viral Bronchitis
	Child (4-13 years)	0-50Hz	Female	Acute Asthma
Crackles	Teenage	0-50Hz	Male	Tuberculosis
	Old (>50 years)	0-50Hz	Female	Rheumatoid
	Child (4-13 years)	250-300Hz	Male	Pneumonia
Stridor	Infant	800-900Hz	Male	Epiglottis
	Infant	650-750Hz	Female	Laryngomalacia

IV. CONCLUSION

All results are concluded after taking at least 5 lung sounds of different people under same conditions. There is very less chance of error in our detection mechanism. The analysis can be made more perfect by taking a greater number of test subjects for analysis. Peaks of the power spectrum have some deviations but all lie in the specified range as mentioned. In this paper, an analysis is carried out to classify the lung sounds to diagnose whether it belongs to an infected person or a normal person. We have successfully developed an algorithm for detection of various diseases given patient's filtered lung sound and his symptoms. The peak of the power spectrum determines the type of disease. We have done analysis for only some of the age groups and diseases due to low availability of filtered lung sounds. Our work can be carried forward by making a full database consisting of many pulmonary diseases for different age groups and genders. Lung sounds of different affected people can be collected, filtered and classified. This standard database can be used as reference and an automated system can be made that detects the disease given the filtered lung sound.

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